

# A Dimensional Scaling Condition for Mixed Gaussian Traces

## Abstract

We study an analytic expression, the *Mixed Gaussian Trace*  $\mathcal{Z}(p, q)$ , defined as the tensor product of the continuous standard Gaussian measure on  $\mathbb{R}^p$  and the discrete heat trace on the flat torus  $\mathbb{T}^q$ , evaluated at the modular self-dual point  $\tau = i$ . By framing this trace within the theory of periods and utilizing the Chowla-Selberg formula, we show that  $\mathcal{Z}(p, q)$  evaluates to a monomial in fractional powers of  $\pi$  and  $\Gamma(1/4)$  over  $\mathbb{Q}(\sqrt{2})$ . Invoking Chudnovsky's theorem on the algebraic independence of  $\pi$  and  $\Gamma(1/4)$ , we establish that the exponent of  $\pi$  exactly vanishes if and only if the dimensions satisfy the scaling relation  $2p = 3q$ . At the fundamental integer solution  $(p = 3, q = 2)$ , the evaluation lies in  $\mathbb{Q}(\sqrt{2}, \Gamma(1/4))$ .

## 1. The Mixed Spectral Trace

We define a mixed-dimensional analytic object via spectral geometry. Let  $\Delta_{\mathbb{T}^q}$  be the positive Laplace-Beltrami operator on the standard flat torus  $\mathbb{T}^q = \mathbb{R}^q/\mathbb{Z}^q$ . The eigenvalues of the Laplacian are  $\lambda_n = 4\pi^2\|n\|^2$  for  $n \in \mathbb{Z}^q$ . The trace of the heat kernel at spectral time  $t > 0$  is given by the multidimensional Jacobi theta function:

$$K_{\mathbb{T}^q}(t) = \text{Tr} \left( e^{-t\Delta_{\mathbb{T}^q}} \right) = \sum_{n \in \mathbb{Z}^q} e^{-4\pi^2\|n\|^2 t} \quad (1)$$

We evaluate this trace at the specific spectral time  $t = \frac{1}{4\pi}$ , corresponding to the self-dual point  $\tau = i$  under the modular transformation  $\tau \mapsto -1/\tau$ . This yields  $K_{\mathbb{T}^q} \left( \frac{1}{4\pi} \right) = \theta_3(e^{-\pi})^q$ . Concurrently, we evaluate the standard unnormalized Gaussian integral over  $\mathbb{R}^p$ .

**Definition 1** (The Mixed Gaussian Trace). *Let  $\mathcal{Z}(p, q)$  be the product of the continuous Gaussian volume of  $\mathbb{R}^p$  and the discrete heat trace on  $\mathbb{T}^q$  at  $t = \frac{1}{4\pi}$ :*

$$\mathcal{Z}(p, q) = \left( \int_{\mathbb{R}^p} e^{-\|x\|^2/2} d^p x \right) \times \text{Tr} \left( e^{-\frac{1}{4\pi}\Delta_{\mathbb{T}^q}} \right) \quad (2)$$

## 2. CM Evaluation and the Period Ring

The continuous integral trivially evaluates to  $(2\pi)^{p/2}$ . The self-dual torus evaluated at  $\tau = i$  corresponds to the elliptic curve  $E : y^2 = x^3 - x$ , which has complex multiplication (CM) by the Gaussian integers  $\mathbb{Z}[i]$ .

By the Chowla-Selberg formula, the periods of CM elliptic curves evaluate analytically in terms of the Euler Gamma function at rational arguments. Specifically, the trace at  $\tau = i$  resolves to:

$$\theta_3(e^{-\pi}) = \frac{\Gamma(1/4)}{2^{1/2}\pi^{3/4}} \quad (3)$$

Taking the product across arbitrary dimensions  $(p, q)$  yields the exact closed form:

$$\mathcal{Z}(p, q) = (2\pi)^{\frac{p}{2}} \cdot \left( \frac{\Gamma(1/4)}{2^{1/2}\pi^{3/4}} \right)^q = 2^{\frac{p-q}{2}} \cdot \pi^{\frac{2p-3q}{4}} \cdot \Gamma\left(\frac{1}{4}\right)^q \quad (4)$$

This expression establishes that  $\mathcal{Z}(p, q)$  evaluates to a monomial in fractional powers of the algebraically independent periods  $\pi$  and  $\Gamma(1/4)$  over the real quadratic field  $\mathbb{Q}(\sqrt{2})$ .

## 3. Exponent Vanishing via Chudnovsky's Theorem

In 1984, G. V. Chudnovsky proved that  $\pi$  and  $\Gamma(1/4)$  are algebraically independent over  $\mathbb{Q}$ . Consequently, no non-trivial polynomial relation exists between them over the algebraic numbers. This dictates that the fractional power of  $\pi$  in Equation 4 cannot be algebraically simplified or absorbed by any rational function of  $\Gamma(1/4)$ .

**Theorem 1** (Vanishing of the  $\pi$ -Exponent). *The exponent of  $\pi$  in the Mixed Gaussian Trace  $\mathcal{Z}(p, q)$  vanishes, so that  $\mathcal{Z}(p, q) \in \mathbb{Q}(\sqrt{2}, \Gamma(1/4))$ , if and only if the dimensional parameters satisfy the linear scaling relation:*

$$2p = 3q \quad (5)$$

## 4. The (3,2) Dimensional Scaling Relation

The minimal positive integer solution satisfying this scaling relation occurs at  $p = 3$  and  $q = 2$ . When evaluated at this dimensional ratio, the exponent of  $\pi$  exactly vanishes ( $\pi^0 = 1$ ):

$$\begin{aligned} \mathcal{Z}(3, 2) &= 2^{\frac{3-2}{2}} \cdot \pi^0 \cdot \Gamma\left(\frac{1}{4}\right)^2 \\ \implies &\boxed{\mathcal{Z}(3, 2) = \sqrt{2} \cdot \Gamma\left(\frac{1}{4}\right)^2 \approx 18.5899 \dots} \end{aligned} \quad (6)$$

**Remark 1.** *This strict dimensional scaling clarifies the structural origin of previously observed heuristic identities. Coupling a 1-dimensional continuous measure with a 2-dimensional discrete trace ( $p = 1, q = 2$ ) yields an invariant strictly proportional to  $\frac{\Gamma(1/4)^2}{\pi\sqrt{2}}$ . The persistence of  $\pi$  in the denominator is the algebraic consequence of a non-zero exponent of  $\pi$ , as the tuple  $(1, 2)$  fails the  $2p = 3q$  condition. Isolating  $\mathcal{Z}(3, 2)$  provides the exact arithmetic scaling where the exponent of  $\pi$  vanishes, yielding an evaluation strictly within the subfield  $\mathbb{Q}(\sqrt{2}, \Gamma(1/4))$ .*